

Weighted BTX: optimized paper tables

Tables 5-8 with fresh dev-box baselines and two-thirds stake profiles

Optimized weighted BTX: paper-table reproduction

At W=1,580 and 16 messages, the fastest measured cold variant is BN254 swapped / 2 at 38.94 ms (5.40x versus the fresh original BLST baseline). Cached BN254 with original groups takes 22.91 ms.

Weight W	Messages M	BLST cold ms	Best cold variant	Best cold ms	Speedup	BN original cached ms
764	16	131.14	BN254 swapped / 2	28.06	4.67x	18.12
1,580	16	210.43	BN254 swapped / 2	38.94	5.40x	22.91
3,063	16	325.41	BN254 swapped / 2	48.21	6.75x	23.72
6,489	16	556.06	BN254 swapped / 2	71.76	7.75x	23.70
1,580	32	378.67	BN254 swapped / 2	58.84	6.44x	37.94
1,580	64	758.11	BN254 original / 2	95.21	7.96x	71.39
1,580	128	1,308.28	BN254 original / 2	180.04	7.27x	155.20
1,580	256	2,604.53	BN254 original / 2	345.77	7.53x	311.57

MEASUREMENT NOTES

All new measurements: AMD EPYC 9275F, 12 workers pinned to physical cores 0-11. Values are arithmetic means, not medians. WBTX: 30 fresh-input samples per cell, two reversed-order rounds, two warmups per case per round.

Cold DecTotal includes all selected local validators' shares, fresh share acceptance, one committee preparation, cross terms and opening. It excludes encryption, client-proof validation, setup and networking. Cached removes only committee preparation.

M is the number of messages processed. Original and swapped BLST use one batch of M; optimized MCL uses M/2 batches of two. Every run retains setup capacity L=M. The committee is the heaviest-first authorized set.

Best cold selects the lowest arithmetic mean among the three optimized MCL variants at that parameter point. This identifies the fastest measured variant in this run, not statistically established dominance. The cached column always shows BN254 with original groups.

Target stake threshold is 2/3. Saved inputs regenerate W=3,063 and 6,489; the PDF prints 3,060 and 6,486. The corresponding q and N match. We use actual generated weights without modifying them to force the printed totals.

These are complete-implementation comparisons. BN254 changes security assumptions; its path has no AVX-512. MCL and BLST differ in validation and target-group arithmetic. WPFE and prior schemes are unmodified references.

Splitting does not improve every case: cached BLS split batches are 15-20% slower than original BLST at 128/256 messages, with no AVX callbacks in those cached runs. BN254 original groups remain the fastest measured cached variant. The cold BN orientation gap at 128 messages is within observed variation.

Weighted BTX: optimized paper tables

Tables 5-8 with fresh dev-box baselines and two-thirds stake profiles

Table 5. Decryption across weight profiles

16 messages; epsilon=1/8, 1/16, 1/32, 1/64. N=234,419,598,688; q=512,1060,2044,4327; selected validators=59,76,79,79.

Weight W	Implementation	PreDec / validator	Validate / validator	DecPrecomp	DecOpen	DecTotal	Cached DecTotal
764	BLST original	0.042	0.165	96.135	22.783	131.142	34.767
764	BLST swapped	0.092	0.130	38.254	25.725	77.118	38.662
764	BLS AVX swapped / 2	0.079	0.123	8.074	14.142	34.164	26.456
764	BN254 swapped / 2	0.045	0.131	7.069	10.596	28.063	21.025
764	BN254 original / 2	0.026	0.103	12.071	10.436	30.139	18.117
764	WPFE (unchanged)	0.039	0.166	88.700	18.245	119.049	30.348
1,580	BLST original	0.041	0.146	169.527	26.671	210.429	40.729
1,580	BLST swapped	0.093	0.113	67.350	29.725	112.750	45.323
1,580	BLS AVX swapped / 2	0.080	0.122	15.500	17.597	48.412	33.128
1,580	BN254 swapped / 2	0.045	0.131	12.388	13.198	38.943	26.578
1,580	BN254 original / 2	0.026	0.102	23.437	13.541	46.734	22.912
1,580	WPFE (unchanged)	0.042	0.146	161.556	22.221	198.055	36.498
3,063	BLST original	0.042	0.143	283.551	27.207	325.412	41.899
3,063	BLST swapped	0.093	0.112	109.171	30.528	155.875	46.573
3,063	BLS AVX swapped / 2	0.082	0.121	26.235	18.757	61.037	34.446
3,063	BN254 swapped / 2	0.046	0.130	20.662	13.649	48.215	27.639
3,063	BN254 original / 2	0.027	0.102	41.495	13.384	65.063	23.720
3,063	WPFE (unchanged)	0.041	0.144	275.744	22.822	313.134	37.389
6,489	BLST original	0.043	0.142	514.297	27.136	556.063	41.781
6,489	BLST swapped	0.096	0.112	193.503	30.435	240.358	46.603
6,489	BLS AVX swapped / 2	0.082	0.120	48.841	18.594	83.456	34.501
6,489	BN254 swapped / 2	0.046	0.129	44.278	13.690	71.764	27.573
6,489	BN254 original / 2	0.027	0.100	82.686	13.826	106.571	23.699
6,489	WPFE (unchanged)	0.042	0.143	508.991	22.813	546.401	37.410

MEASUREMENT NOTES

Times in ms. PreDec and Validate are pooled wall times divided by the number of selected real validators, across all chunks. They are amortized rates, not a single validator's standalone latency. DecPrecomp is committee preparation; DecOpen is cross-term precomputation plus opening.

WBTX DecTotal is directly timed. WPFE uses 30 online repetitions over two encrypted-batch fixtures, and its comparable total is the sum of rounded phases excluding client checks. Its cached value subtracts preparation; it is not a separately timed cached run. Fresh WBTX ciphertexts and proofs change every iteration.

The paper does not fully specify its timing denominators or sample count. Its MacBook M4 measurements are preserved in paper_reference.json; this report's speedups use only fresh measurements on the dev box.

Weighted BTX: optimized paper tables

Tables 5-8 with fresh dev-box baselines and two-thirds stake profiles

Table 6. Decryption across message counts

W=1,580; N=419; q=1,060; 76 selected validators with accepted weight 1,062. All messages are recovered in each sample.

Messages M	Implementation	PreDec / validator	Validate / validator	DecPrecomp	DecOpen	DecTotal	Cached DecTotal
16	BLST original	0.041	0.146	169.527	26.671	210.429	40.729
16	BLST swapped	0.093	0.113	67.350	29.725	112.750	45.323
16	BLS AVX swapped / 2	0.080	0.122	15.500	17.597	48.412	33.128
16	BN254 swapped / 2	0.045	0.131	12.388	13.198	38.943	26.578
16	BN254 original / 2	0.026	0.102	23.437	13.541	46.734	22.912
16	WPFE (unchanged)	0.042	0.146	161.556	22.221	198.055	36.498
32	BLST original	0.065	0.221	307.425	49.485	378.675	70.671
32	BLST swapped	0.189	0.156	118.768	57.687	202.632	83.215
32	BLS AVX swapped / 2	0.159	0.226	15.609	27.280	72.166	56.710
32	BN254 swapped / 2	0.090	0.250	12.481	20.455	58.842	46.500
32	BN254 original / 2	0.051	0.182	23.718	20.534	61.970	37.938
32	WPFE (unchanged)	0.065	0.221	286.099	38.506	346.346	60.246
64	BLST original	0.104	0.412	609.870	109.001	758.107	147.526
64	BLST swapped	0.304	0.252	233.961	128.389	404.654	170.057
64	BLS AVX swapped / 2	0.319	0.366	15.599	53.978	121.659	106.730
64	BN254 swapped / 2	0.174	0.393	12.563	40.364	96.014	83.638
64	BN254 original / 2	0.103	0.307	23.748	40.309	95.210	71.387
64	WPFE (unchanged)	0.106	0.416	575.426	84.604	699.722	124.296
128	BLST original	0.174	0.781	1,117.570	118.108	1,308.283	192.182
128	BLST swapped	0.511	0.438	420.180	124.354	616.604	196.922
128	BLS AVX swapped / 2	0.625	0.822	15.674	110.781	236.464	221.610
128	BN254 swapped / 2	0.348	0.802	12.478	81.970	181.872	170.704
128	BN254 original / 2	0.204	0.746	23.909	83.949	180.038	155.200
128	WPFE (unchanged)	0.174	0.765	1,105.992	174.221	1,351.553	245.561
256	BLST original	0.295	1.502	2,228.659	239.283	2,604.530	376.998
256	BLST swapped	0.874	0.821	836.282	251.959	1,217.097	379.788
256	BLS AVX swapped / 2	1.256	1.726	15.583	227.486	469.708	453.344
256	BN254 swapped / 2	0.697	1.813	12.468	174.481	377.696	376.665
256	BN254 original / 2	0.401	1.400	23.844	185.011	345.771	311.572
256	WPFE (unchanged)	0.296	1.507	2,198.399	234.316	2,569.742	371.343

MEASUREMENT NOTES

Times in ms. PreDec and Validate are pooled wall times divided by the number of selected real validators, across all chunks. They are amortized rates, not a single validator's standalone latency. DecPrecomp is committee preparation; DecOpen is cross-term precomputation plus opening.

WBTX DecTotal is directly timed. WPFE uses 30 online repetitions over two encrypted-batch fixtures, and its comparable total is the sum of rounded phases excluding client checks. Its cached value subtracts preparation; it is not a separately timed cached run. Fresh WBTX ciphertexts and proofs change every iteration.

The paper does not fully specify its timing denominators or sample count. Its MacBook M4 measurements are preserved in paper_reference.json; this report's speedups use only fresh measurements on the dev box.

Weighted BTX: optimized paper tables

Tables 5-8 with fresh dev-box baselines and two-thirds stake profiles

Table 7a. Encryption and keys across weights

16 messages; setup capacity L=16 for every implementation.

Weight W	Implementation	Enc ms/item	Core dk KiB	Verify keys KiB	Total key KiB	Share bytes / validator
764	BLST original	0.069	2,220.4	351.0	2,571.4	48
764	BLST swapped	0.083	1,110.2	175.5	1,285.7	96
764	BLS AVX swapped / 2	0.069	1,110.2	175.5	1,285.7	768
764	BN254 swapped / 2	0.047	740.1	117.0	857.1	512
764	BN254 original / 2	0.044	1,480.2	234.0	1,714.2	256
764	WPFE (unchanged)	0.068	2,292.1	351.0	2,643.1	48
1,580	BLST original	0.069	4,591.9	628.5	5,220.4	48
1,580	BLST swapped	0.083	2,295.9	314.2	2,610.2	96
1,580	BLS AVX swapped / 2	0.068	2,295.9	314.2	2,610.2	768
1,580	BN254 swapped / 2	0.048	1,530.6	209.5	1,740.1	512
1,580	BN254 original / 2	0.044	3,061.2	419.0	3,480.2	256
1,580	WPFE (unchanged)	0.083	4,740.1	628.5	5,368.6	48
3,063	BLST original	0.069	8,901.8	897.0	9,798.8	48
3,063	BLST swapped	0.083	4,450.9	448.5	4,899.4	96
3,063	BLS AVX swapped / 2	0.069	4,450.9	448.5	4,899.4	768
3,063	BN254 swapped / 2	0.048	2,967.3	299.0	3,266.3	512
3,063	BN254 original / 2	0.044	5,934.6	598.0	6,532.6	256
3,063	WPFE (unchanged)	0.079	9,189.1	897.0	10,086.1	48
6,489	BLST original	0.070	18,858.7	1,032.0	19,890.7	48
6,489	BLST swapped	0.083	9,429.3	516.0	9,945.3	96
6,489	BLS AVX swapped / 2	0.068	9,429.3	516.0	9,945.3	768
6,489	BN254 swapped / 2	0.047	6,286.2	344.0	6,630.2	512
6,489	BN254 original / 2	0.043	12,572.4	688.0	13,260.4	256
6,489	WPFE (unchanged)	0.093	19,467.1	1,032.0	20,499.1	48

MEASUREMENT NOTES

Enc is pooled encryption plus proof-generation wall time divided by M. WBTX values use cold-mode fresh samples. WPFE encrypts once per process, so its Enc mean has two observations, not 30.

Key sizes are exact compressed-group payload counts in KiB (1024 bytes), matching the paper's arithmetic despite its kB label. Core WBTX points=(2L-1)W; verification points=N*L; total is their sum. The encryption key, proof CRS, metadata and wire framing are excluded, as in Table 7.

Every timed run retains setup L=M. Public point widths are 96/48 bytes for BLS original/swapped and 64/32 bytes for BN254 original/swapped. Share widths are the opposite source group: 48/96 and 32/64 bytes. The last column includes all M/2 shares for a split run.

Weighted BTX: optimized paper tables

Tables 5-8 with fresh dev-box baselines and two-thirds stake profiles

Table 7b. Encryption and keys across message counts

W=1,580; setup capacity L=M is retained even when the working chunk is two.

Messages M	Implementation	Enc ms/item	Core dk KiB	Verify keys KiB	Total key KiB	Share bytes / validator
16	BLST original	0.069	4,591.9	628.5	5,220.4	48
16	BLST swapped	0.083	2,295.9	314.2	2,610.2	96
16	BLS AVX swapped / 2	0.068	2,295.9	314.2	2,610.2	768
16	BN254 swapped / 2	0.048	1,530.6	209.5	1,740.1	512
16	BN254 original / 2	0.044	3,061.2	419.0	3,480.2	256
16	WPFE (unchanged)	0.083	4,740.1	628.5	5,368.6	48
32	BLST original	0.053	9,331.9	1,257.0	10,588.9	48
32	BLST swapped	0.064	4,665.9	628.5	5,294.4	96
32	BLS AVX swapped / 2	0.052	4,665.9	628.5	5,294.4	1,536
32	BN254 swapped / 2	0.036	3,110.6	419.0	3,529.6	1,024
32	BN254 original / 2	0.033	6,221.2	838.0	7,059.2	512
32	WPFE (unchanged)	0.052	9,480.1	1,257.0	10,737.1	48
64	BLST original	0.056	18,811.9	2,514.0	21,325.9	48
64	BLST swapped	0.067	9,405.9	1,257.0	10,662.9	96
64	BLS AVX swapped / 2	0.051	9,405.9	1,257.0	10,662.9	3,072
64	BN254 swapped / 2	0.035	6,270.6	838.0	7,108.6	2,048
64	BN254 original / 2	0.033	12,541.2	1,676.0	14,217.2	1,024
64	WPFE (unchanged)	0.050	18,960.1	2,514.0	21,474.1	48
128	BLST original	0.050	37,771.9	5,028.0	42,799.9	48
128	BLST swapped	0.061	18,885.9	2,514.0	21,399.9	96
128	BLS AVX swapped / 2	0.051	18,885.9	2,514.0	21,399.9	6,144
128	BN254 swapped / 2	0.035	12,590.6	1,676.0	14,266.6	4,096
128	BN254 original / 2	0.032	25,181.2	3,352.0	28,533.2	2,048
128	WPFE (unchanged)	0.050	37,920.1	5,028.0	42,948.1	48
256	BLST original	0.049	75,691.9	10,056.0	85,747.9	48
256	BLST swapped	0.059	37,845.9	5,028.0	42,873.9	96
256	BLS AVX swapped / 2	0.051	37,845.9	5,028.0	42,873.9	12,288
256	BN254 swapped / 2	0.035	25,230.6	3,352.0	28,582.6	8,192
256	BN254 original / 2	0.032	50,461.2	6,704.0	57,165.2	4,096
256	WPFE (unchanged)	0.047	75,840.1	10,056.0	85,896.1	48

MEASUREMENT NOTES

Enc is pooled encryption plus proof-generation wall time divided by M. WBTX values use cold-mode fresh samples. WPFE encrypts once per process, so its Enc mean has two observations, not 30.

Key sizes are exact compressed-group payload counts in KiB (1024 bytes), matching the paper's arithmetic despite its kB label. Core WBTX points=(2L-1)W; verification points=N*L; total is their sum. The encryption key, proof CRS, metadata and wire framing are excluded, as in Table 7.

Every timed run retains setup L=M. Public point widths are 96/48 bytes for BLS original/swapped and 64/32 bytes for BN254 original/swapped. Share widths are the opposite source group: 48/96 and 32/64 bytes. The last column includes all M/2 shares for a split run.

Weighted BTX: optimized paper tables

Tables 5-8 with fresh dev-box baselines and two-thirds stake profiles

Table 8. Prior-scheme reference measurements

Unmodified implementations on the same 12-core dev-box allocation; M=16. Ten measured repetitions per point across two rounds.

Weight W	Scheme	Enc ms/item	PreDec ms / denominator	Source core Dec ms	With checks ms	PreDec denominator
764	PFE	0.082	0.011	21.142	24.189	per item / virtual server
764	BTX	0.526	0.011	20.649	24.143	per item / virtual server
764	BEAT++	0.088	0.015	87.463	87.941	per real validator
1,580	PFE	0.082	0.011	55.825	58.813	per item / virtual server
1,580	BTX	0.528	0.011	55.099	58.575	per item / virtual server
1,580	BEAT++	0.088	0.014	145.197	145.677	per real validator
3,063	PFE	0.082	0.011	173.893	176.861	per item / virtual server
3,063	BTX	0.527	0.011	172.751	176.165	per item / virtual server
3,063	BEAT++	0.088	0.014	223.056	223.533	per real validator
6,489	PFE	0.082	0.011	733.241	736.190	per item / virtual server
6,489	BTX	0.528	0.011	728.740	732.151	per item / virtual server
6,489	BEAT++	0.127	0.014	367.822	368.302	per real validator

MEASUREMENT NOTES

These rows preserve the upstream runners' distinct meanings. BTX/PFE use q virtual parties for the selected set, generate their shares serially, divide by q for a one-server average, then divide by M for PreDec/item. Their core total includes that one-server average, precomputation, combination and opening; With checks adds client and combined-share checks.

BEAT++ PreDec divides pooled share-generation time by all N real validators. Its core total includes all-N shares and response checks, selected-committee preparation, cross terms and opening. With checks also includes client-proof validation. These totals are not an identical distributed workload and should not be used for unqualified cross-scheme speedups.

BTX/PFE use fresh full runs with one unmeasured warmup per process. BEAT++ uses ten independent one-run processes without an upstream warmup. Raw logs preserve its threshold/security diagnostic for this two-thirds configuration. These are performance observations, not a security claim.

Weighted BTX: optimized paper tables

Tables 5-8 with fresh dev-box baselines and two-thirds stake profiles

Full online pipeline and deployment tradeoffs

Encryption and client-proof validation are included in these additional end-to-end measurements.

Weight W	Messages M	BLST cold ms	BN swapped cold ms	BN original cached ms	BN swapped L=2 key KiB	BN swapped share bytes / validator
764	16	132.59	29.76	19.53	86.2	512
1,580	16	211.88	40.63	24.31	174.3	512
3,063	16	326.86	49.90	25.14	324.5	512
6,489	16	557.52	73.43	25.11	651.3	512
1,580	32	380.91	61.41	40.06	174.3	1,024
1,580	64	762.74	100.99	75.54	174.3	2,048
1,580	128	1,316.73	191.73	163.51	174.3	4,096
1,580	256	2,620.94	397.88	328.42	174.3	8,192

MEASUREMENT NOTES

The online timer includes fresh encryption/proofs, client validation, all local validator shares, acceptance, optional committee preparation, cross terms and all openings. Setup and networking are excluded; all plaintexts are checked after every warmup and measured iteration.

The L=2 key column is a calculated deployment option, not the setup used for the timings: constrain the setup itself to chunks of two, giving (3W+2N) compressed public-key points. This key is independent of M but cannot support a single batch larger than two.

Smaller chunks increase responses: M/2 source-group elements per validator instead of one. The table reports raw group payload only. Cached measurements require the same setup, chunk size and accepted committee; all shares and verification are fresh.

Measured curves: BLS12-381 and MCL BN_SNARK1 (Ethereum BN254). AVX-512 is BLS G1 only and is verified through actual callback counters. The MCL protocol is an experimental trusted-dealer, in-memory implementation with no network/wire decoder.